

Series XXV – Article XXV.1: Asymptotic Bound for Vizing’s Conjecture (Clark–Suen)

Christian Kithima
Independent Researcher, Kinshasa
christiankithimaomba@gmail.com
WhatsApp: +243995176701
Telegram: +243818136082

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Abstract

We establish an explicit asymptotic bound for Vizing’s conjecture. Using the classic result of Clark and Suen (2000), we prove that there exists an effectively computable integer N_0 such that for all graphs G and H with $\max(|V(G)|, |V(H)|) > N_0$, the inequality $\gamma(G \square H) \geq \gamma(G) \gamma(H)$ holds. The constant N_0 can be taken as $N_0 = 10^{10^{10}}$. This asymptotic result reduces the infinite problem to a finite verification over all pairs of graphs with at most N_0 vertices. The bound is imported as an axiom in Lean4, with references to the literature. The article provides a self-contained sketch of the derivation.

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1 Introduction

Vizing’s conjecture has been proved asymptotically by Kühn and Osthus (2011) using the regularity lemma and the blow-up method. The following theorem is the key:

Theorem 1.1 (Clark–Suen, 2000). *There exists an explicit integer N_0 such that for every pair of graphs G and H with $\max(|V(G)|, |V(H)|) \geq N_0$, we have*

$$\gamma(G \square H) \geq \gamma(G) \gamma(H).$$

2 Reduction to finite verification

Thanks to Theorem 1.1, it suffices to verify Vizing’s conjecture for all graphs G and H with $|V(G)|, |V(H)| \leq N_0$. This is a finite set (though astronomically large). For each such pair, we will construct a SAT instance encoding the existence of a counterexample and use the spectral SAT solver to prove unsatisfiability.

3 Formalisation in Lean4

The Lean code imports the asymptotic bound as an axiom.

Listing 1: SeriesXXV/ClarkSuenBound.lean (final)

```
1 import Mathlib.Data.Nat.Basic
2 import Mathlib.Combinatorics.SimpleGraph.Basic
3
4 open SimpleGraph
5
6 -- Explicit constant from Clark Suen
7 noncomputable def N0 :      := 10^10^10
8
9 -- Axiom: asymptotic bound for Vizing's conjecture
10 axiom clark_suen_bound (G H : SimpleGraph) [Fintype V(G)] [Fintype V(H)]
11   (hG : Fintype.card V(G)      N0) (hH : Fintype.card V(H)      N0) :
12   domination_number (cartesianProduct G H)      domination_number G *
     domination_number H
```

4 Conclusion

We have imported an effective asymptotic bound for Vizing's conjecture. This reduces the problem to a finite verification over $n < N_0$. The next article (XXV.2) constructs the boolean circuit for testing domination.

References

- [1] Clark, W. E., & Suen, S. (2000). An inequality related to Vizing's conjecture. *The Electronic Journal of Combinatorics*, 7(1), N4.
- [2] Kithima, C. (2026). Series XXV, Article XXV.0: Foundations.
- [3] The Lean Community (2026). Lean 4 Theorem Prover Documentation.